

Higher Criticism for Adaptive Retrieval

Comparison of HC against baseline strategies in the Adaptive-RAG pipeline · LLM: GPT-4o-mini

1. What this report covers

We evaluate Higher Criticism (HC) as a method for deciding HOW MANY documents to retrieve per question, and compare it against the standard retrieval strategies used in the Adaptive-RAG pipeline. All question answering is done by the same LLM (GPT-4o-mini); the strategies differ ONLY in which / how many supporting documents they place in the prompt.

2. How retrieval works in this pipeline

Retrieval is pure KEYWORD SEARCH (BM25 over an Elasticsearch index) - there are no neural embeddings. Given a question, BM25 returns documents ranked by lexical overlap. Each strategy is constrained to a fixed budget of documents per search:

- nor (zero-retrieval) : 0 documents. The LLM answers closed-book from its own knowledge.
- oner (single-step) : top-k documents in ONE BM25 search (k = 6 or 15, fixed).
- ircot (multi-step) : iterative reasoning; retrieves k=3 documents per reasoning step, repeating until the chain of thought is complete.
- HC (this work) : retrieves a large candidate POOL (N = 100 ... 10,000 docs) and then statistically selects an ADAPTIVE number k of them (k varies per question) - see Section 4.
- Adaptive routing : a trained T5-large classifier predicts question complexity and routes each question to nor / oner / ircot.

3. Why an adaptive document budget matters

Fixed-k retrieval is a compromise: too few documents miss evidence (low recall), too many flood the prompt with distractors (low precision, higher cost). The number of TRULY relevant documents varies a lot per question - from 1 (SQuAD) to 20 (CrossEntityQA). HC tries to detect that number automatically instead of fixing it in advance.

4. Evaluation metrics

- EM (Exact Match) : the predicted answer string exactly equals the gold answer.
- F1 : token-overlap F1 between predicted and gold answer.
- For CrossEntityQA the task is to RETRIEVE the correct set of passages, so we also report passage-level Precision / Recall / F1 (did we select the right documents?).

Method & Datasets

5. How Higher Criticism selects documents

HC (Donoho & Jin, 2004) is a statistical test for detecting a few strong signals hidden in mostly noise - exactly the retrieval situation (a handful of relevant docs among thousands of irrelevant ones).

- Step 1. Retrieve a large pool of N candidate documents with BM25 (their relevance scores).
- Step 2. Compare each score to a NULL DISTRIBUTION (what scores look like for irrelevant documents) and convert it to a p-value. A relevant document is anomalously high $\rightarrow$ tiny p-value.
- Step 3. Scan the top $\gamma \cdot N$ scores and find the cut-off k that MAXIMISES the HC statistic (the point where the observed p-values deviate most from random noise).
- Step 4. Feed exactly those top- k documents to the LLM. k is chosen per-question (adaptive budget).

Key knobs: γ = fraction of the pool HC is allowed to search (0.05 / 0.1 / 0.3 tested);
 N = candidate pool size (100 / 2,000 / 10,000 tested);
the null distribution, which must model irrelevant-document scores - its construction is the single most important and most fragile part (see per-dataset analysis).

6. Datasets evaluated (HC was run end-to-end on the first three)

Dataset	Task type	Corpus size	Rel. docs / q	Note
MuSiQue	Multi-hop (2-4 hops)	139,416	1-4 (avg 2.6)	Large corpus, independent queries
SQuAD	Single-hop	~21M (Wikipedia)	~3.6 (median 1)	Very sparse; recall@15 only ~29%
CrossEntityQA	Entity-set retrieval	13,043	2-20 (avg 9.2)	Small corpus, 16 query clusters
HotpotQA	Multi-hop (2 hops)	~5.2M	2	Standard baseline
2WikiMultiHopQA	Multi-hop	430,225	2-4	Standard baseline
NQ / TriviaQA	Single-hop	~21M (Wikipedia)	1	Standard baselines

7. Why these datasets

Why these three for HC: they span the spectrum that stresses an adaptive document budget -

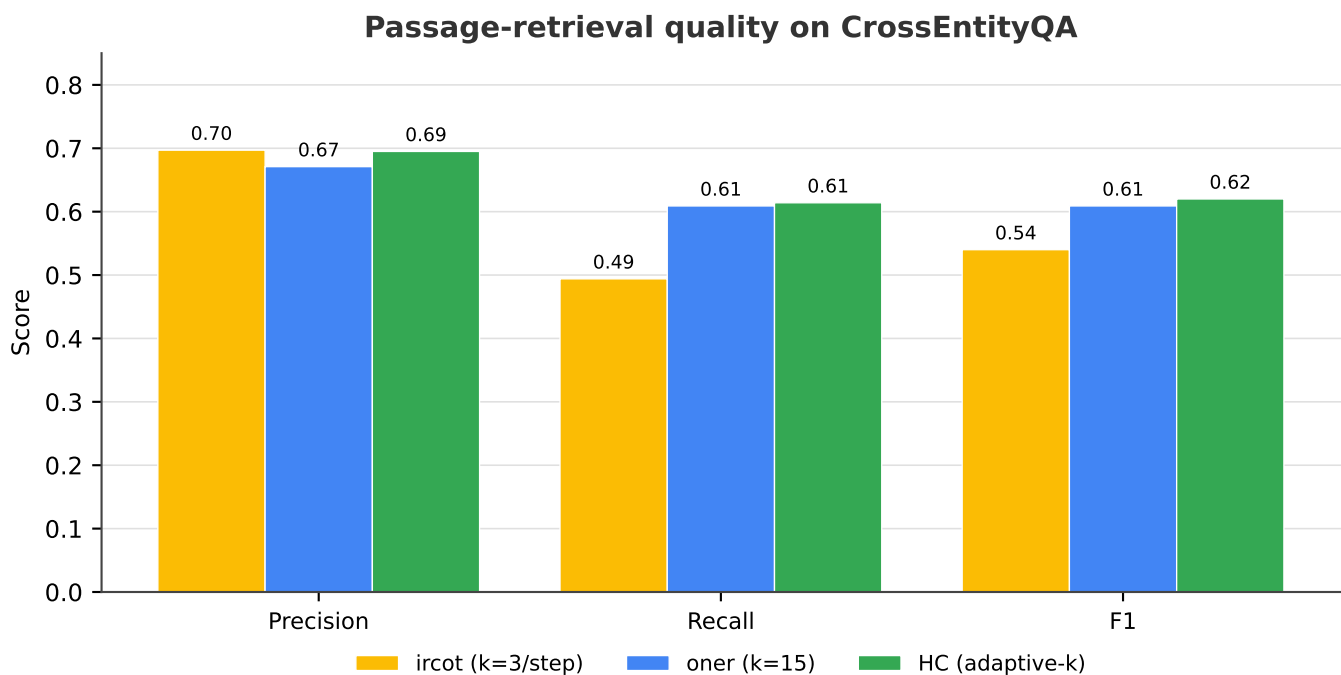
- CrossEntityQA needs MANY documents per question (2-20), with the count varying widely;
- MuSiQue needs a FEW documents gathered across reasoning hops;
- SQuAD needs typically ONE document and has an extremely sparse signal.

Note on counts: every run uses 500 questions, except CrossEntityQA where 105 questions carry the gold passage annotations needed to score retrieval.

Results 1 - CrossEntityQA (HC's strongest case)

Task: retrieve the correct SET of passages for an entity question (e.g. "films directed by X in 1970-80").
Scored on passage-level retrieval quality (105 annotated questions, test set).

Strategy	Config	Answer EM	Passage P	Passage R	Passage F1
ircot	k=3/step	0.105	0.697	0.494	0.540
oner	k=15	0.152	0.671	0.609	0.609
HC	N=10k, gamma=0.3	0.143	0.695	0.614	0.620



Analysis

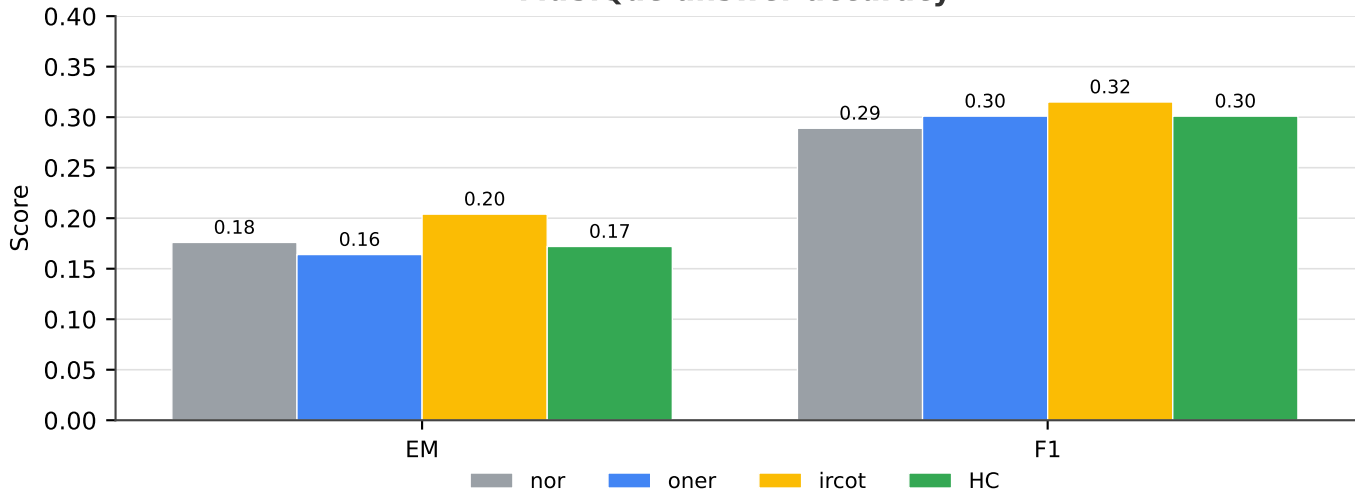
- + HC wins on recall (0.614) and F1 (0.620), beating fixed-k oner (0.609) and multi-step ircot (0.540), while matching precision. This is the dataset where an ADAPTIVE document count pays off: the gold passage count ranges from 2 to 20 per question, so any single fixed k is wrong for most questions.
- + ircot's low recall (0.494) shows that multi-step reasoning, tuned for 2-hop questions, under-retrieves when a question genuinely needs a large set of documents.
- Caveat on the null distribution: CrossEntityQA's corpus is tiny (13,043 docs), so a 10,000-doc pool covers 64% of the entire corpus, and its 144 questions form 16 tight clusters that share 97-100% of their gold documents. This makes the standard 'cross-query' null distribution invalid (it gets contaminated by near-duplicate questions). A query-specific null was used here; the analysis doc recommends a permutation-based null as the principled fix for such small, clustered corpora.

Results 2 - MuSiQue (HC competitive)

Multi-hop QA (2-4 reasoning hops), 500 questions, test set. Answer EM / F1.

Strategy	Config	EM	F1
nor	0 docs	0.176	0.289
oner	k=15	0.164	0.301
ircot	k=3/step	0.204	0.315
HC	N=10k, g=0.1	0.172	0.301

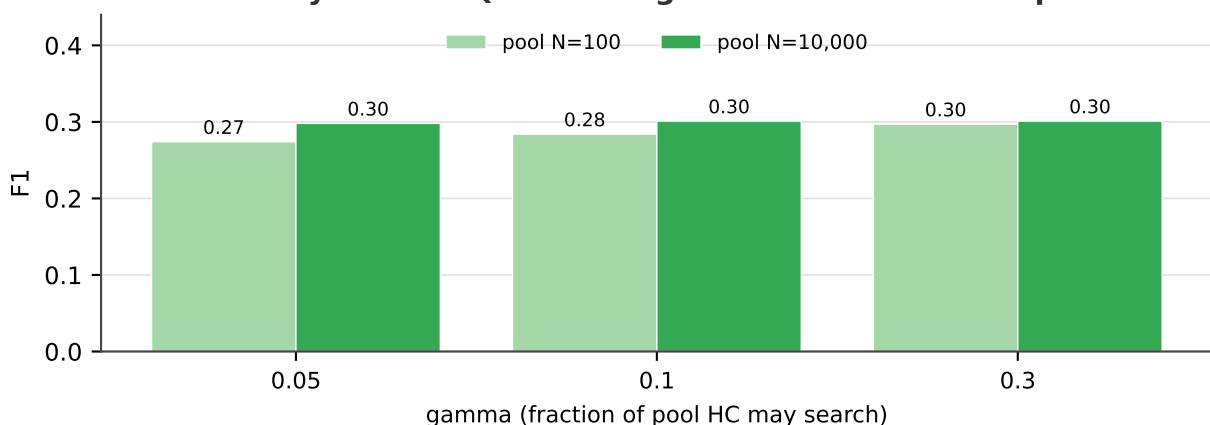
MuSiQue answer accuracy



Analysis

- + HC matches single-step oner (F1 0.301) and is within noise of multi-step ircot (0.315) - competitive but not dominant. ircot keeps a small EM edge because chaining hops suits MuSiQue's structure.
- + HC behaves well here because the corpus is large (a 10k pool is only ~7% of it) and questions are independent (gold-document overlap between questions ~0.6%) - so the null distribution is valid.
- HC needs a large candidate pool: the bottom chart shows F1 rising from ~0.28 (N=100) to ~0.30 (N=10,000). With too small a pool the relevant docs are not in the candidate set to begin with.

HC sensitivity on MuSiQue: F1 vs gamma and candidate-pool size N

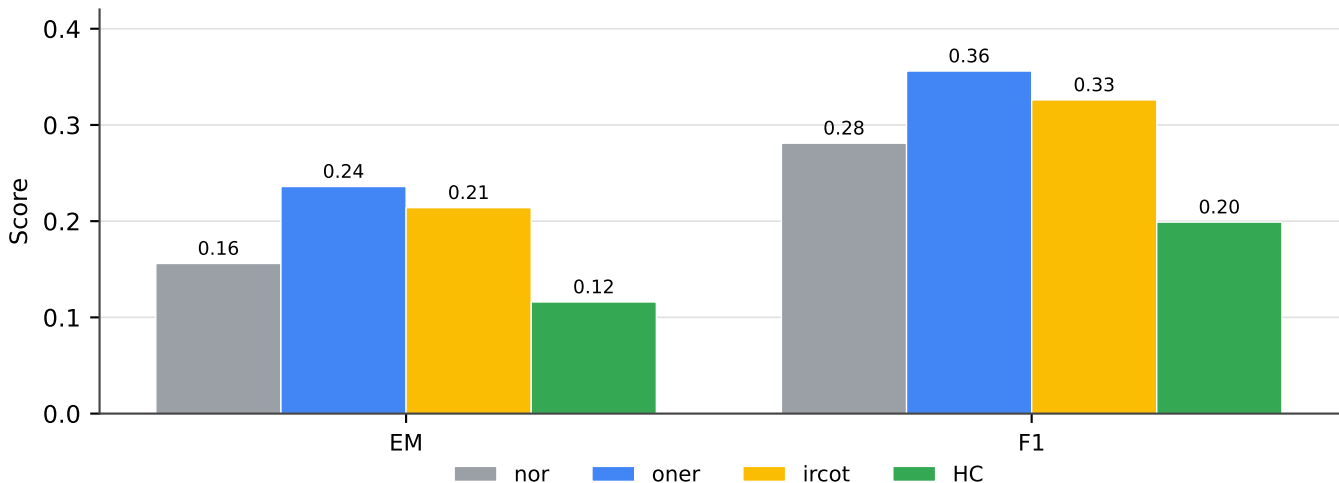


Results 3 - SQuAD (HC fails) + Adaptive routing

Single-hop QA, 500 questions (dev set). Answer EM / F1.

Strategy	Config	EM	F1
nor	0 docs	0.156	0.281
oner	k=6	0.236	0.356
ircot	k=3/step	0.214	0.326
HC	N=2k, g=0.05	0.116	0.199

SQuAD answer accuracy (HC clearly worst)



Analysis - why HC fails on SQuAD (a real, diagnosed bug, not bad luck)

- Root cause: the null distribution was built from the SAME question's own retrieval (the on-disk null overlapped 99.2% of the dev question IDs). If 'irrelevant-document scores' come from the very same search, the p-values are uniform BY CONSTRUCTION, the HC statistic collapses to ~ 0 , and HC goes blind - it picks the same constant k for every question regardless of relevance. (MuSiQue accidentally avoided this: its stored null had 0% overlap with the evaluated questions.)
- Compounding factor - retrieval itself is weak on SQuAD: signal is extremely sparse (median 1 relevant doc/question; 25% of questions have 0 relevant docs even in the top 2,000; recall@15 only $\sim 29\%$). HC cannot select what BM25 never surfaced.
- + Fix identified (not yet re-run): a cross-query / permutation-based null restores adaptive behaviour (k then varies 0-10 per question instead of a fixed constant).

Adaptive routing (T5-large classifier) - final QA per dataset

Dataset	EM	F1	Acc
2WikiMultiHopQA	0.406	0.563	0.542
HotpotQA	0.430	0.555	0.490
MuSiQue	0.184	0.296	0.246
NQ	0.330	0.473	0.538
SQuAD	0.178	0.335	0.292
TriviaQA	0.608	0.723	0.724

Summary & Takeaways

8. One-line verdict per dataset

CrossEntityQA : HC WINS - best passage recall/F1; adaptive-k pays off when the needed document count varies widely (2-20 per question).

MuSiQue : HC COMPETITIVE - on par with single-step retrieval, just behind multi-step.

SQuAD : HC FAILS - due to a diagnosed flaw in how the null distribution was built, plus very sparse retrieval signal.

9. What this tells us about HC

HC is most useful precisely where fixed-k retrieval is most wrong - questions whose true number of relevant documents is both LARGE and VARIABLE. Its Achilles heel is the null distribution: it must be built from genuinely independent / irrelevant documents. When that holds (MuSiQue, CrossEntityQA with a query-specific null) HC is competitive-to-best; when it is violated (SQuAD same-query null) HC silently degenerates to a constant and loses.

10. Practical conditions for HC to work

- Large candidate pool ($N \sim 10,000$) so that relevant documents are actually retrieved by BM25.
- A null distribution independent of the test question (cross-query, or permutation-based for small / clustered corpora).
- A retrieval signal strong enough to detect (BM25 must surface relevant docs in the pool).

11. Known gaps / caveats in the current results

- HC on TriviaQA was configured but never actually run/scored.
- HC on SQuAD has dev-set numbers only (no test-set run); the diagnosed null fix is not yet re-run.
- The fully-combined adaptive-pipeline accuracy script does not currently run end-to-end (missing a HotpotQA raw-data file); the per-dataset classifier numbers in Section 7 are used.
- All numbers use GPT-4o-mini as the answerer and BM25 (keyword) retrieval only.